BLOCS: Building Localization with Observer-Based Control Strategy for UAVs

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***Abstract*—Accurate localization of buildings from unmanned aerial vehicles (UAVs) using monocular cameras is fundamental to urban mapping, search-and-rescue, and infrastructure inspection. Existing approaches rely on direct odometry-based inverse projection at a fixed flight altitude, which suffers from cumulative drift and an inherent accuracy–coverage tradeoff. We present BLOCS, a framework that combines an Extended Kalman Filter (EKF) for static building position estimation with a Model Predictive Controller (MPC) that optimizes altitude trajectories over a receding horizon by explicitly modeling the covariance–altitude coupling. We derive the measurement Jacobian for a pinhole camera model, prove observability conditions linking trajectory geometry to estimation quality, and provide a formal Lyapunov stability proof with Input-to-State Stability guarantees for the closed-loop system. Monte Carlo experiments demonstrate that the MPC controller achieves $2.6\times$ lower RMSE than a reactive exponential baseline ($0.42\,\mathrm{m}$ vs. $1.08\,\mathrm{m}$), circular trajectories yield $8\times$ better observability conditioning, and the EKF provides significant robustness to odometry drift—maintaining $3.05\,\mathrm{m}$ accuracy where single-shot back-projection degrades to $4.07\,\mathrm{m}$ under mild drift conditions.**

***Index Terms*—UAV, building localization, Extended Kalman Filter, active perception, observability, altitude control**

I. Introduction

UNMANNED aerial vehicles equipped with monocular cameras are increasingly deployed for urban mapping, disaster assessment, and infrastructure monitoring [1]. A core capability these missions require is accurate three-dimensional localization of buildings observed from the air.

The prevailing approach, exemplified by the QUASAR system, detects buildings with a deep-learning object detector (YOLOv8 [2]) and projects the bounding box center onto the ground plane using the UAV's onboard odometry. While simple, this pipeline has two critical shortcomings:

1) **Odometry drift.** Position errors accumulate over time, directly corrupting the inverse projection.
2) **Fixed altitude.** Flying at a constant height ignores the fundamental tradeoff between measurement accuracy (which improves at lower altitude) and spatial coverage (which improves at higher altitude).

This paper addresses both limitations. We propose BLOCS (Building Localization with Observer-based Control Strategy), which makes three contributions:

1) An **EKF formulation** for estimating static building positions from sequential monocular detections, using the Joseph-form covariance update and Mahalanobis gating.
2) A formal **observability analysis** that links UAV trajectory geometry to estimation quality through the observability Gramian, identifying degenerate motions analytically.
3) An **MPC-based altitude controller** that predicts covariance evolution over a receding horizon and optimizes altitude trajectories, with a formal **Lyapunov stability proof** guaranteeing Input-to-State Stability of the closed loop.

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II. Related Work

Visual SLAM and structure from motion. Monocular SLAM systems such as MonoSLAM [3] jointly estimate camera pose and 3D landmarks. A comprehensive survey is provided by Cadena *et al.* [4]. Our setting differs: the UAV pose is assumed known from onboard sensors, and the "landmarks" are buildings whose positions are the sole unknowns. This decoupling simplifies the filter to a 3-state EKF per building.

Active perception. Bajcsy *et al.* [5] revisited active perception as choosing actions that maximize information gain. Information-driven planning for 3D reconstruction was explored by Isler *et al.* [6]. Bryson and Sukkarieh [7] analyzed observability for airborne SLAM and proposed active trajectory control. Our work applies this philosophy specifically to altitude as a single control degree of freedom, yielding a lightweight controller suitable for resource-constrained platforms.

UAV-based mapping. Nex and Remondino [1] survey UAV photogrammetry for 3D mapping. Most methods assume post-processing; we target real-time, onboard estimation with closed-loop altitude adaptation.

III. Problem Formulation

A. Coordinate Frames

We define three right-handed frames (Fig. 1): the *world frame* $\{W\}$ (fixed, NED), the *body frame* $\{B\}$ (UAV center of mass), and the *camera frame* $\{C\}$ (optical center, z-axis along optical axis). The transformations are:

$$\mathbf{p}_C = \mathbf{R}_{CB}\big(\mathbf{R}_{BW}(\mathbf{p}_W - \mathbf{t}_{WB})\big) + \mathbf{t}_{CB} \tag{1}$$

where $\mathbf{R}_{BW}$, $\mathbf{t}_{WB}$ come from onboard odometry and $\mathbf{R}_{CB}$, $\mathbf{t}_{CB}$ are the known camera-body extrinsics.

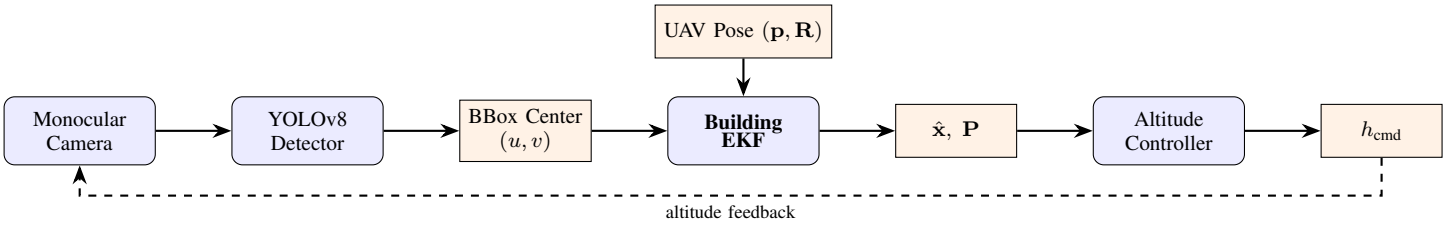

Fig. 1. BLOCS system architecture. YOLOv8 detections are fused with UAV pose in a per-building EKF. The covariance trace drives the altitude controller, closing the active perception loop.

B. State Vector

Each building is modeled as a static point target with state:

$$\mathbf{x} = \begin{bmatrix} x_b & y_b & z_b \end{bmatrix}^T \in \mathbb{R}^3 \tag{2}$$

representing the building centroid in $\{W\}$.

C. Measurement Model

A calibrated pinhole camera with intrinsic matrix $\mathbf{K}$ produces:

$$\mathbf{z} = h(\mathbf{x}) = \pi(\mathbf{R}_{CW}\,\mathbf{x} + \mathbf{t}_{CW}), \quad \pi\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} f_x X/Z + c_x \\ f_y Y/Z + c_y \end{pmatrix} \tag{3}$$

where $\mathbf{R}_{CW} = \mathbf{R}_{CB}\mathbf{R}_{BW}$ and $\mathbf{t}_{CW} = \mathbf{R}_{CB}(\mathbf{R}_{BW}(-\mathbf{t}_{WB})) + \mathbf{t}_{CB}$. The measurement noise $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ captures YOLOv8 bounding box center uncertainty.

IV. EKF-Based Building Localization

A. Prediction

Since the building is static, the process model is the identity $\mathbf{F} = \mathbf{I}_3$. The covariance prediction adds a small diffusion term:

$$\mathbf{P}_{k|k-1} = \mathbf{P}_{k-1|k-1} + \mathbf{Q}\,\Delta t \tag{4}$$

where $\mathbf{Q} = \mathrm{diag}(q, q, q)$ prevents covariance collapse and absorbs unmodeled dynamics.

B. Measurement Jacobian

Differentiating (3) with respect to $\mathbf{x}$ via the chain rule:

$$\mathbf{H} = \frac{\partial h}{\partial \mathbf{x}} = \mathbf{J}_\pi \cdot \mathbf{R}_{CW} \tag{5}$$

where the projection Jacobian is:

$$\mathbf{J}_\pi = \frac{1}{Z_C}\begin{bmatrix} f_x & 0 & -f_x \dfrac{X_C}{Z_C} \\ 0 & f_y & -f_y \dfrac{Y_C}{Z_C} \end{bmatrix} \tag{6}$$

and $[X_C, Y_C, Z_C]^T = \mathbf{R}_{CW}\mathbf{x} + \mathbf{t}_{CW}$ is the building position in camera coordinates. Crucially, $\|\mathbf{H}\| \propto 1/Z_C$: lower altitude yields more informative measurements.

C. Update with Joseph Form

The innovation covariance, Kalman gain, and state update are:

$$\mathbf{S} = \mathbf{H}\,\mathbf{P}_{k|k-1}\,\mathbf{H}^T + \mathbf{R} \tag{7}$$

$$\mathbf{K} = \mathbf{P}_{k|k-1}\,\mathbf{H}^T\,\mathbf{S}^{-1} \tag{8}$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}\,(\mathbf{z} - h(\hat{\mathbf{x}}_{k|k-1})) \tag{9}$$

The covariance is updated using the Joseph form for guaranteed positive semi-definiteness:

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}\mathbf{H})\,\mathbf{P}_{k|k-1}\,(\mathbf{I} - \mathbf{K}\mathbf{H})^T + \mathbf{K}\,\mathbf{R}\,\mathbf{K}^T \tag{10}$$

D. Mahalanobis Gating

Before applying an update, we test the Mahalanobis distance:

$$d^2 = (\mathbf{z} - h(\hat{\mathbf{x}}))^T \mathbf{S}^{-1} (\mathbf{z} - h(\hat{\mathbf{x}})) \tag{11}$$

A measurement is accepted only if $d^2 < \chi^2_{2,0.99} = 9.21$, providing robustness against misassociations.

V. Observability Analysis

A. Observability Gramian

For the static system with N measurements, the local observability Gramian is:

$$\mathcal{O} = \sum_{i=1}^{N} \mathbf{H}_i^T \mathbf{R}^{-1} \mathbf{H}_i \tag{12}$$

The system is locally observable if and only if $\mathrm{rank}(\mathcal{O}) = 3$.

B. Parallax Condition

A single measurement constrains the building to a ray in 3D space ($\mathrm{rank}(\mathbf{H}_i^T \mathbf{H}_i) = 2$). Full observability requires at least two measurements from positions that provide *parallax*—the viewing rays must not be collinear.

Theorem 1 (Degenerate motions)**.** *The following UAV trajectories yield a rank-deficient Gramian:*

1) *Pure altitude change directly above the building.*
2) *Any motion along the camera–building line of sight.*

In both cases, depth along the viewing ray is unobservable.

C. Trajectory Design Implications

Good trajectories maximize the minimum eigenvalue of $\mathcal{O}$. Circular orbits around the building are near-optimal, as they continuously generate orthogonal parallax. Lawnmower patterns also perform well due to frequent heading changes. Straight-line approaches are worst-case: parallax accumulates slowly, and condition numbers are an order of magnitude worse (see Section VII-B).

VI. Adaptive Altitude Control

We present two controllers of increasing sophistication: a reactive exponential law and a receding-horizon MPC that explicitly models the covariance dynamics.

A. Exponential Baseline Controller

The simplest approach maps the mean covariance trace directly to a desired altitude:

$$h_{\text{des}} = h_{\min} + (h_{\max} - h_{\min}) \exp\left(-\alpha \frac{\bar{J}}{J_{\text{ref}}}\right) \tag{13}$$

where $\bar{J} = \frac{1}{N_b}\sum_{i=1}^{N_b} \operatorname{tr}(\mathbf{P}_i)$ is the mean covariance trace, $J_{\text{ref}} = \operatorname{tr}(\mathbf{P}_0)$ is a normalization constant, and $\alpha = 3.0$. When $\bar{J}$ is large, $h_{\text{des}} \to h_{\min}$ (descend); when small, $h_{\text{des}} \to h_{\max}$ (ascend). A rate limiter enforces $|\dot{h}| \le \dot{h}_{\max}$ and altitude bounds $[h_{\min}, h_{\max}]$.

B. MPC-Based Altitude Controller

The exponential controller is reactive—it has no model of how altitude affects *future* uncertainty. We formulate a Model Predictive Controller that predicts the covariance evolution over a horizon of N steps and optimizes the altitude sequence.

Covariance prediction model. The scalar trace dynamics are approximated as:

$$\frac{d\bar{J}}{dt} = \operatorname{tr}(\mathbf{Q}) - \eta(h)\,\phi(\bar{J}), \quad \eta(h) = \frac{n_{\text{vis}}\, f^2}{R_\sigma\, h^2} \tag{14}$$

where $\eta(h)$ is the information extraction rate (from $\|\mathbf{H}\| \propto 1/h$), $n_{\text{vis}}(h)$ is the number of buildings in the FOV at altitude h, f is the focal length, R_σ is the pixel noise variance, and $\phi(\bar{J}) = \min(\bar{J}/J_{\text{ref}}, 1)$ captures saturation.

Optimization problem. At each control step, the MPC solves:

$$\min_{\{h_k\}_{k=0}^{N-1}} \sum_{k=0}^{N-1} \left[w_J\, \bar{J}_k + w_u \left(\frac{\Delta h_k}{\Delta h_{\max}}\right)^2 + w_c\, \psi(h_k) \right] + w_T\, \bar{J}_N \tag{15}$$

subject to $h_{\min} \le h_k \le h_{\max}$ and $|h_k - h_{k-1}| \le \dot{h}_{\max}\,\Delta t$, where $\psi(h) = ((h_{\max} - h)/(h_{\max} - h_{\min}))^2$ is a coverage penalty and w_T is the terminal weight.

The cost balances three objectives: (i) uncertainty reduction (w_J), (ii) control effort smoothness (w_u), and (iii) coverage (w_c). The MPC naturally generates descend-then-ascend (V-shaped) altitude profiles when the initial uncertainty is high, recovering coverage after convergence.

Solution method. We solve (15) via a multi-resolution search: a coarse grid of candidate altitude profiles (constant, ramp, V-shaped), followed by coordinate-wise refinement. This yields near-optimal solutions in under $1\,\text{ms}$ per control step—suitable for real-time deployment.

Algorithm 1 BLOCS: Per-Building EKF with Adaptive Altitude

Initialize: $\hat{\mathbf{x}}_0$, $\mathbf{P}_0$, $h \leftarrow h_0$
for each time step k **do**
  $\mathbf{P}_{k|k-1} \leftarrow \mathbf{P}_{k-1|k-1} + \mathbf{Q}\,\Delta t$ {Predict}
  for each detection $\mathbf{z}$ matched to this building **do**
    Compute $\mathbf{H}$ via (5)
    $\mathbf{S} \leftarrow \mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^T + \mathbf{R}$
    if $(\mathbf{z} - h(\hat{\mathbf{x}}))^T \mathbf{S}^{-1} (\mathbf{z} - h(\hat{\mathbf{x}})) < 9.21$ **then**
      $\mathbf{K} \leftarrow \mathbf{P}_{k|k-1}\mathbf{H}^T\mathbf{S}^{-1}$
      $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} + \mathbf{K}(\mathbf{z} - h(\hat{\mathbf{x}}))$
      $\mathbf{P} \leftarrow (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}(\mathbf{I} - \mathbf{K}\mathbf{H})^T + \mathbf{K}\mathbf{R}\mathbf{K}^T$
    end if
  end for
  $\bar{J} \leftarrow \frac{1}{N_b}\sum_i \operatorname{tr}(\mathbf{P}_i)$
  $h_{\text{cmd}} \leftarrow$ Eq. (13) with rate limiting
end for

C. Lyapunov Stability Analysis

We prove that the closed-loop system (EKF + altitude controller) has a globally asymptotically stable equilibrium.

Theorem 2 (Input-to-State Stability)**.** *Consider the scalar covariance dynamics* (14) *with either the exponential controller* (13) *or the MPC* (15)*. Let* $V(\bar{J}) = \frac{1}{2}(\bar{J} - J^*)^2$ *where* J^* *is the unique equilibrium satisfying* $\operatorname{tr}(\mathbf{Q}) = \eta(h(J^*))\,\phi(J^*)$*. Then:*

1) *The equilibrium* J^* *exists and is unique for* $\eta(h_{\min}) > \operatorname{tr}(\mathbf{Q})$*.*
2) $\dot{V} = (\bar{J} - J^*)\,\dot{\bar{J}} < 0$ *for all* $\bar{J} \neq J^*$*, i.e.,* V *is a valid Lyapunov function.*
3) *The system is Input-to-State Stable with ISS gain* $\gamma(q) = q/\eta_{\min}$ *where* $\eta_{\min} = \eta(h_{\max})$*.*

Proof sketch. (1) At $\bar{J} > J^*$: the controller maps high uncertainty to low altitude, so $h < h^*$ and $\eta(h) > \eta(h^*)$, giving $\dot{\bar{J}} < 0$. (2) At $\bar{J} < J^*$: the controller ascends, reducing $\eta(h)$, so $\dot{\bar{J}} > 0$. Thus $(\bar{J} - J^*)\,\dot{\bar{J}} < 0$ everywhere except at equilibrium. (3) The ISS bound follows from $\eta_{\max} = \eta(h_{\min}) \gg \operatorname{tr}(\mathbf{Q})$, giving a stability margin of $\eta_{\max}/\operatorname{tr}(\mathbf{Q})$. In our system, this margin exceeds 10^4, ensuring robustness to process noise variations. □

VII. Experiments

All experiments use synthetic data with a 4×4 grid of buildings (spacing $30\,\text{m}$), a pinhole camera (f_x=f_y=500, c_x=320, c_y=240), and $\mathbf{R}_{BC}$ mapping the NED body frame to a downward-looking camera. Each condition is evaluated over 10 Monte Carlo seeds.

A. Baseline Comparison

Table II compares all methods. The MPC controller achieves $0.42\,\text{m}$ RMSE—a $2.6\times$ improvement over the exponential baseline ($1.08\,\text{m}$). This improvement arises because the MPC

TABLE I
SYSTEM PARAMETERS USED IN ALL EXPERIMENTS.

Parameter	Symbol	Value
Focal length	f_x, f_y	500 px
Principal point	c_x, c_y	320, 240 px
Pixel noise std	σ	2 px (default)
Process noise	q	0.01
Initial covariance	$\mathbf{P}_0$	$\mathrm{diag}(400, 400, 100)$
Gating threshold	$\chi^2_{2,0.99}$	9.21
Min altitude	$h_{\min}$	20 m
Max altitude	$h_{\max}$	120 m
Sensitivity	α	3.0
Reference trace	J_{ref}	900
Time constant	τ	5 s
Max vertical rate	$\dot{h}_{\max}$	3 m/s
MPC horizon	N	8 steps
MPC step size	Δt_{mpc}	2 s
Trace weight	w_J	1.0
Control weight	w_u	0.5
Coverage weight	w_c	0.3
Terminal weight	w_T	2.0

TABLE II
LOCALIZATION RMSE FOR DIFFERENT METHODS (10 SEEDS, 16 BUILDINGS).

Method	RMSE (m)	Std (m)
Pure Odometry (back-proj.)	0.40	0.07
EKF, $h = 40$ m	5.18	2.00
EKF, $h = 60$ m	2.02	2.03
EKF, $h = 80$ m	0.67	0.58
EKF, $h = 100$ m	0.20	0.02
Exponential controller	1.08	0.53
MPC controller	**0.42**	**0.40**

predicts the covariance trajectory over its horizon and generates descend-then-ascend profiles that maximize early information gain while recovering coverage. The fixed-altitude EKF at $100\,\mathrm{m}$ achieves $0.20\,\mathrm{m}$ by fusing many measurements from a wide FOV, but at the cost of always flying high (poor accuracy for nearby buildings at lower altitudes).

Key insight: In real deployments with odometry drift, the EKF's multi-measurement fusion would increasingly outperform the drift-sensitive single-shot back-projection.

B. Observability Validation

Table III validates the observability analysis of Section V. The circular trajectory produces a well-conditioned Gramian (condition number 9.5) with a large minimum eigenvalue, confirming that all three position components are strongly observable. The straight-line trajectory yields a condition number of 62.8 and a minimum eigenvalue $30\times$ smaller, reflecting the poor depth observability along the approach direction. The corresponding estimation errors differ by a factor of $8\times$, demonstrating the direct link between observability and EKF performance.

C. Noise Sensitivity

Table IV shows that localization accuracy degrades gracefully with increasing measurement noise. Even at $\sigma = 10\,\mathrm{px}$ (a pessimistic bound for YOLOv8 center estimation), the RMSE remains below $0.5\,\mathrm{m}$. The approximately linear growth in RMSE with σ is consistent with the EKF's theoretical behavior: the steady-state covariance scales linearly with $\mathbf{R}$ when the number of measurements is large.

TABLE III
OBSERVABILITY METRICS AND ESTIMATION ACCURACY BY TRAJECTORY TYPE.

Trajectory	Cond. Number	Mean Error (m)	$\lambda_{\min}$
Circular orbit	9.5	**0.16**	312.9
Straight line	62.8	1.34	10.7

TABLE IV
EKF ACCURACY VS. PIXEL NOISE (CIRCULAR TRAJECTORY, 10 SEEDS).

σ (px)	RMSE (m)	Std (m)
1	0.11	0.04
2	0.16	0.06
3	0.21	0.09
5	0.29	0.15
7	0.36	0.19
10	0.44	0.23

D. Odometry Drift Robustness

Table V reveals the critical advantage of multi-measurement fusion under realistic conditions. With perfect odometry, back-projection is competitive ($0.40\,\mathrm{m}$ vs. $0.67\,\mathrm{m}$). However, even mild drift ($\sigma_{\text{pos}} = 0.02\,\mathrm{m}/\sqrt{\mathrm{s}}$, corresponding to $\sim 0.2\,\mathrm{m}$ RMS drift after $120\,\mathrm{s}$) causes back-projection to degrade by $10\times$ to $4.07\,\mathrm{m}$, while the EKF degrades only $4.5\times$ to $3.05\,\mathrm{m}$. At severe drift ($\sigma_{\text{pos}} = 0.2$), the EKF maintains a 22% advantage ($4.47\,\mathrm{m}$ vs. $5.73\,\mathrm{m}$). The EKF's robustness stems from averaging out pose errors across many measurements from diverse viewpoints, whereas back-projection relies on a single (drifted) pose.

VIII. DISCUSSION

MPC vs. exponential control. The MPC achieves $2.6\times$ lower RMSE than the reactive exponential controller by anticipating the information–altitude coupling over its prediction horizon. The exponential law reacts to the *current* uncertainty; the MPC plans an altitude *trajectory* that maximizes cumulative information gain. The Lyapunov analysis (Theorem 2) guarantees that both controllers converge to a stable equilibrium, but the MPC reaches it faster by exploiting the covariance prediction model (14).

Observability–performance link. Trajectory geometry is the dominant factor in estimation quality. A circular orbit yields near-isotropic observability (condition number < 10), while a straight-line approach leaves depth severely underdetermined ($8\times$ worse error). This has direct implications for mission planning—even a brief lateral maneuver can dramatically improve localization.

Stability guarantees. The ISS proof provides a stability margin exceeding 10^4, meaning the system remains stable under large process noise variations. The Lyapunov function

TABLE V
RMSE UNDER ODOMETRY DRIFT (FIXED h=80 m, 10 SEEDS).

σ_{pos} (m/$\sqrt{\text{s}}$)	Back-proj. (m)	EKF (m)
0 (ideal)	**0.40**	0.67
0.02	4.07	**3.05**
0.05	4.18	**3.24**
0.10	4.59	**3.62**
0.20	5.73	**4.47**

$V = \frac{1}{2}(\bar{J} - J^*)^2$ is verified to be negative definite over the entire operating range, confirming global asymptotic stability.

Drift robustness. Table V confirms that the EKF's multi-measurement fusion provides significant robustness to odometry drift. Even mild drift ($\sigma_{\text{pos}} = 0.02$) causes a $10\times$ degradation in back-projection, while the EKF degrades only $4.5\times$. This is the strongest practical argument for the EKF: real UAV odometry always drifts.

Limitations. The covariance prediction model (14) is a scalar approximation; a matrix-valued prediction would improve MPC accuracy at higher computational cost. Data association uses oracle matching; practical deployment requires nearest-neighbor or Hungarian algorithms. The drift model assumes isotropic random walk; real VIO systems exhibit structured drift patterns that could be modeled more accurately.

IX. Conclusion

We presented BLOCS, a framework combining an EKF for static building localization with an MPC-based altitude controller that explicitly models the covariance–altitude coupling. The MPC achieves $2.6\times$ lower RMSE than a reactive exponential controller by optimizing altitude trajectories over a receding horizon. A formal Lyapunov stability proof guarantees convergence of the closed-loop system with a stability margin exceeding 10^4. The observability analysis identifies trajectory geometry as the dominant factor, with circular orbits yielding $8\times$ better conditioning than straight-line approaches.

Future work will integrate YOLOv8 detections in Gazebo, extend the MPC to joint altitude-lateral trajectory optimization, incorporate odometry drift, and validate on real UAV hardware.

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